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Problem

Colour-positive film yield pleasing colours, but many are discontinued. Emulation offers an alternative, typically relying on either:

(1) Look-up tables (LUTs). Created manually or from film-digital pairs, they are prone to interpolation artefacts and lack interpretability.

(2) Neural networks. Require many image pairs for training; this data is costly to obtain on film.

(3) Proprietary methods. Not accessible.

We aim to address all of these limitations.

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Proposed Method

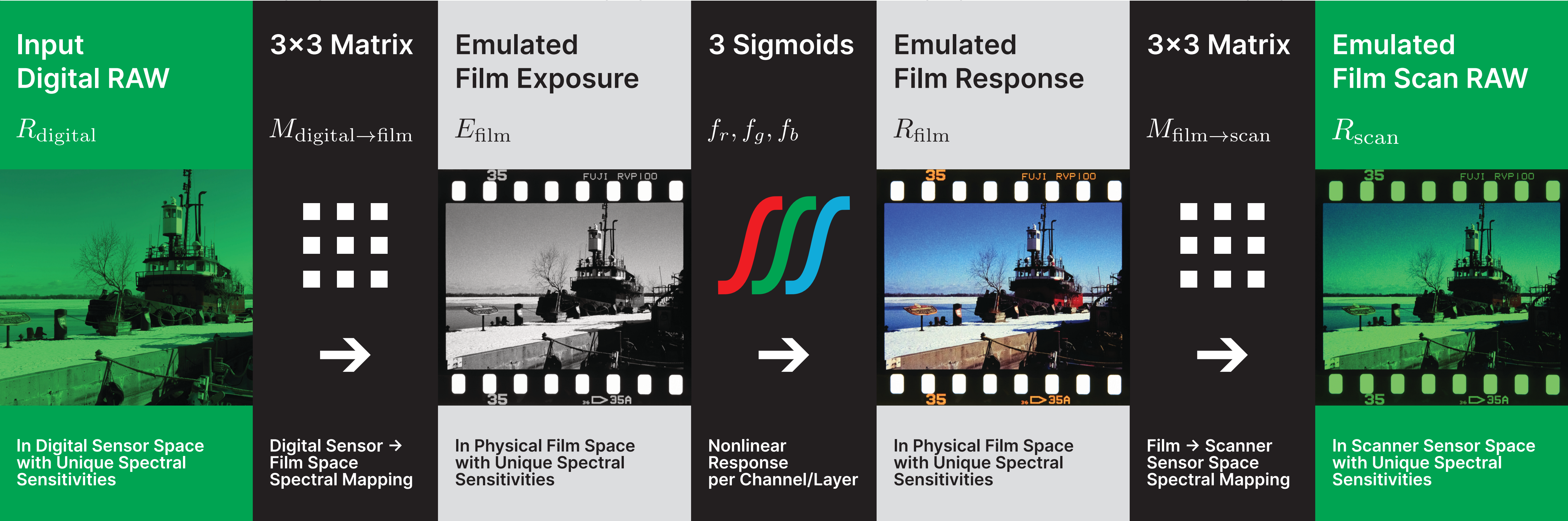
We map a RAW image R_{digital} to an emulated RAW image of scanned film R_{scan} , as a three-stage pipeline mirroring real film processes:

(1) Spectral remapping $M_{\text{digital} \rightarrow \text{film}}$. A 3×3 matrix converts R_{digital} into the energy that would have reached the layers of the target film E_{film} .

(2) Three sigmoid response functions f_r, f_g, f_b . Applied channel (layer)-wise to E_{film} , yielding the film's optical density response R_{film} .

(3) Scanner encoding $M_{\text{film} \rightarrow \text{scan}}$. A second 3×3 matrix converts R_{film} , illuminated by a (virtual) backlight, into the scanner's RAW space R_{scan} .

This yields 30 parameters, optimised via least-squares using film-digital colour pairs captured from a single roll of film.



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Results

Real Film Scan RAW



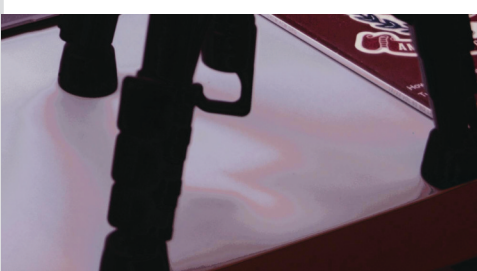
All RAW images (scan, input, and emulation output) were rendered in sRGB for easy viewing and comparison, using fixed, consistent transformations.

*Rendered with official emulation software by the film manufacturer

Input Digital RAW
(of Same Scene)



LUT Baseline
Banding Artefacts

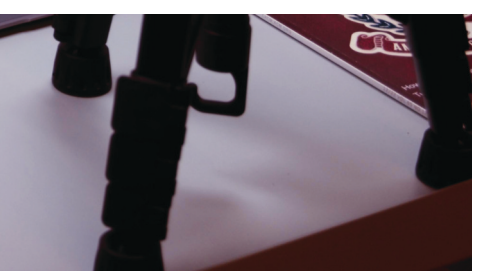


200% Zoom
on Smooth
Area

Proprietary*
Inaccurate Colour



Our Method
Accurate Emulation



200% Zoom
on Smooth
Area

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Applications

- (1) "Film look" in colour grading.** Because the emulation is continuous, we can generate LUTs of any resolution, which can be dropped seamlessly into existing production pipelines.
- (2) Archival preservation.** Artists can digitise the palette of an out-of-production film with a single existing roll, confident that every parameter maps to a verifiable physical quantity.
- (3) Creative authoring.** Vendors may expose parameters as sliders (e.g. "shadow falloff"), enabling artists to manipulate the emulation in ways that are still physically viable.

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Further Work

An in-depth look, other projects, art, and more, via QR:

